

**MI&CT ESERVICE NOTE
SN07062814b**

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This short-term informational note is invalid 6 months after the issue date.

Date: June 13, 2007

Subject: LFC changes for 07MW18.x SW (LS7.X), (16 May 2007) and features list

Effectivity: VCT (GOC5) 7.x Systems

This information should be kept with the 07MW18.x software set.

Affected Service Publication : 5193575-800 - LS 7.X Limited Access Service Info (Adv), Rev 2

PQR/SPR #: FCTge29419, FCTge29668

Features Overview:

- **Reconstruction Speed:** for the time-to-first image: has increased slightly for some modes due to implementation of changes for IQ improvements. Many modes are unchanged. Reconstruction speed and time-to-first image depends on the scan type, slice thickness and pitch selected. Thicker slices will have longer recon times per image compared to 0.625mm slices. Images acquired in the Head, Ped Head, Small Head SFOV will have increased recon times due to the application of IBO (iterative bone option). Cardiac image reconstruction does not start until all data has been acquired for the scan.
- **Reduced Max mA if Tube Warm-Up is Bypassed:** limitations are placed on the maximum mA allowed for the small and large focal spots in the exam following a skipped or cancelled Tube warm-Up. Although the mA will be limited to 250mA for the small focal spot and 500mA for large focal spot, the user can achieve adequate mAs by adjusting the rotation time for the exam. If Tube-Warm-Up is done, mA capabilities will be restored to full range. This protects the tube from selecting high mA when the tube is cold. Messages will be displayed in the OC message area as well as a pink pop-up describing the mA limitations.
- **AutomA/SmartmA:** changes have been implemented to provide more consistent behavior across varied patient size. The maximum Noise Index limit has been increased from 50 to 70. Adjustments in Noise Index may need to be made to adjust the noise level now seen in the images. AutomA/SmartmA is automatically turned off if the series orientation does not match the scout orientation. The system will provide a pop-up indicating AutomA has been turned off. AutomA/SmartmA can be turned on again once the series orientation matches the Scout.

- **Patient Schedule Update:** revised to provide selective update of the Patient Schedule. User can now select the following Update Parameters: Get Patient List for, Date Range, Requested Procedure ID, Accession Number, Patient name or Patient ID on the same pop-up menu.
- **Preset Delay for Auto Voice:** provides the capability to add a preset delay time before x-ray on for patients who may have difficulty holding their breath. Or it could be used in cardiac imaging to increase time for the heart rate to stabilize after a breath hold. The range for the preset delay for Auto Voice is 0 to 7 seconds.
- **SmartPrep:** there have been a number of changes in software to improve reliability of SmartPrep. AutoScan mode is automatically enabled whenever SmartPrep is selected for an acquisition. Message is displayed in console information bar indicating AutoScan is enabled if it was off. All IGs are locked for SmartPrep even though 1 IG is used to reconstruct Monitor Scan images. AutoView is cancelled for series in the recon queue when a SmartPrep series is started. Images from these series can be selected from the Browser using List Select. The images from the series with SmartPrep will be the next series seen in AutoView.

AutoScan mode is automatically enabled whenever SmartPrep is selected for an acquisition. Message displayed in console information bar indicating AutoScan is enabled if it was off.

- **Contrast Presets:** IV and GI (Oral) contrast information can now be entered directly or selected from preset files for IV and Oral contrast. The preset list of contrast information is saved and restored as part of System State.
- **Auto Transfer by Series:** Enhanced to include Scouts, Dose Report, PMR auto Transfer for Recon 1, 2 and 3. Up to 4 hosts may be selected for data transfer.
- **Prospective Multi-Recon (PMR):** PMR now supports Cardiac scan modes for SnapShot Segment, SnapShot Burst, and SnapShot Segment Plus. The Phase % will be enabled in PMR for Recon 2 and Recon 3 for Cardiac modes. Enter the Phase Range and Phase Increment desired as you would in Retro Recon. SnapShot Segment Plus can be selected as a Recon mode for Recon 2 and Recon 3. PMR is not available for SnapShot Pulse or Volume (Axial) Shuttle scan types.
- **Iterative Bone Option (IBO) and Axial IQ Improvements:** changes have been made for better visualization of structures at the bone/brain interface and the Axial and Cine Full IQ.
Axial mode for 40mm beam supports 0.625 and 1.25mm slice thickness reconstruction in Retro Recon for LightSpeed VCT/VCT XT and 1.25 mm slice thickness reconstruction in Retro Recon for LightSpeed Pro³²/VCT Select.
- **Large SFOV 3D Scatter Correction Improvements:** improvements in 3D Scatter correction for Large SFOV to eliminate streak artifact at the level of the shoulders on small patients scanned with the Large SFOV.
- **Bone Plus and Edge Reconstruction Algorithms:** improved to reduce aliasing artifacts seen when using these reconstruction algorithms (kernels) and 0.5x:1 pitch.

- **Patient Orientation – AutomA:** if patient orientation is different from the protocol, AutomA will be disabled. Once the orientation matches, AutomA can be enabled again.
- **Direct VIS – Auto Applications:** Direct VIS column on the ViewEdit screen is now called Auto Apps. Auto Apps supports the selection of Neuro 3D Filter option and Direct VIS options for direct MPR and Direct 3D.
- **Priority Recon:** has been disabled for some acquisition modes where IQ Enhance or SnapShot Pulse has been prescribed for the series. This was done to improve efficiency of recon of these features.
- **Series Names for Reformats:** can be specified in Reformat or Volume Viewer when creating Batch reconstructions. Select Filming Tools and Film/Save options and enable Name Batch Series. Once Name Batch Series has been enabled, you will be able to enter a protocol name for the reformat series being created or use the default name for the protocol as the series description.
- **Large Font Annotation:** has been enhanced to support scan date for display on screen or film (set in User Pref.). The Large Font Annotation information is saved and restored as part of System State.
- **Save on Screen Changes:** this allows the user to define the series description for 3D images.
- **Exam/Series Text Page:** selected from the control panel in Viewer on the Image Works desktop can be screen saved to create a secondary capture of these pages for networking or archiving to desired locations.
- **IQ Enhance:** is a special reconstruction mode for helical scans that can only be selected for 0.62 mm and 1.25mm slice thicknesses to minimize image artifacts (pinwheel) commonly seen in thin slice helical acquisitions. Time-to-first image and image-to-image times will increase slightly for acquisitions when IQ Enhance is enabled due to the additional processing for this feature.

Images will be annotated with an E next to the algorithm and recon mode on the left side of the image to indicate IQ Enhance algorithm was applied.

IQ Enhance is not compatible with Axial or Cine scan types or cardiac gated acquisitions.

IQ Enhance is selected on the Recon tab under Recon Options. If the slice thickness or interval is not valid, IQ Enhance button will be insensitive. The intervals equal to the slice thickness or one half of the slice thickness supported for IQ Enhance. For 0.625mm slice thickness, intervals of 0.625mm and 0.312 are valid for IQ Enhance. For 1.25mm slice thickness, intervals of 1.25mm and 0.625 are valid for IQ Enhance.

New Purchasable Options:

- **Neuro 3D Filter Option:** This is a noise reduction filter optimized for thin slice images of the head that are going to be used for 3D processing. Neuro 3D Filter maintains image sharpness while reducing noise in the image. Reduction of noise can be used to reduce dose for an acquisition or improve IQ.
- **Auto Filter and Transfer – Option:** Auto Filter and Transfer allows user to prospectively filter images at the Low/N1, Medium/N2 or High/N3 setting for the Neuro 3D Filter and network the filtered images to desired location. **Neuro 3D Filter Option is needed for this feature to work.** Neuro 3D Filter is not compatible with DMPR or Direct3D.
- **Volume (Axial) Shuttle Mode – Option:** Special Axial scan mode where the cradle shuttles back and forth between two adjacent 40mm beam collimations, proving total coverage of 80mm for Neuro CT Perfusion acquisitions. Tilt is not allowed.
- **SnapShot Pulse – Option:** This is a prospective gated segment cine mode with a single sector reconstruction method for cardiac imaging. PMR is not supported for this mode. SnapShot Pulse is compatible for *stable* heart rates up to 65 BPM. System automatically selects a padding value based on heart rate to support variation in heart rate during acquisition. Padding is added to both sides of the acquisition window. Padding range is 0 to 200 msec. Dose reduction compare to Cardiac Helical modes. Amount depends on heart rate and padding for the acquisition. This Cardiac Cine mode provides a dose savings up to 70% compared to Cardiac Helical modes.

New Service Features:

- **Image Artifact IQ Snap:**
This function is designed to capture core system/scan/recon configuration and usage information at the time an Image Quality problem is found at a customer system. Select the images with IQ issues that you want to reserve the scan data files for in the Browser on Image Works desktop and make an anonymous patient by image. Select Utilities in the Browser and create an Anonymous Patient by image, select the images within the Anonymous Exam and then select IQ Snap in the Tool Bar in Image Works. Select OK in the IQ Snap Warning pop-up to start the process and then select OK in the IQ Snap pop-up to reserve the scan data files. A pop-up will be displayed at the end of the save indicating if the save was successful or not. Do not initiate an IQ Snap while the system is actively scanning.
The data captured at the time of the artifact can be loaded onto a DVD and shipped to HQ for analysis.
- **Cardiac RTS:**

This function is designed to gather run time statistics (RTS) for Cardiac exams. This data collected during applications scanning, or aborts, does not affect system performance.

- **Gantry Accuracy Improvement:**

The gantry accuracy has been improved from 0.25 degrees to 0.14 degrees at zero degree tilt position via a firmware changed. This helps to reduce the Stair Stepping artifact.

Hardware Related Issues:

None

Software Load Issues:

SL #1: VDARC Node Boot BIOS Order does not get changed after the software load is successful.

Issue: Incorrect Note in Section 7.2 Operating System Software Installation on Host Computer for VDARC.

If issues occur during the VDARC OS load, a pop-up may appear stating that the VDARC BIOS must be checked. If the VDARC load continues to fail, determine if the problem is the Ethernet cable, Host NIC, or VDARC Node.

After saving the VDARC BIOS, the *BIOS SETUP UTILITY* window should disappear (within approximately one minute) and the VDARC will begin to reboot. If this window does not disappear, close the window and attempt to load the VDARC Node. The VDARC Node may require a reset or re-power.

NOTE: Issues have been reported by sites using 06MW29.7 software with Jarrell hardware. In each case, the Save was successful; however, if the load does fail, re-perform the VDARC BIOS Setup Utility process.

~~If the DARC BIOS is changed, it must be returned to original settings after the VDARC Node software load is complete for proper normal operation. Failure to return the VDARC Node Boot Order BIOS after the load process is complete may result in VDARC boot issues.~~

Solution: End of Note should be removed because the VDARC Node (Westville and Jarrell) does not require any additional changes for VDARC Boot Order BIOS setup if the VDARC software has loaded successfully.

SL #2: Section 9 CT Applications Start Up may halt the Operator Console

Issue: Operator Console may halt when Application software is started due to an unknown reason. This has been seen on a VCT with xw8000 Host and Westville VDARC Node hardware.

Solution: After the 'System halted' message appears power off the VCT Operator Console at the front switch. Wait 30 seconds for the disk drives to settle. Apply power to the VCT Operator Console at the front switch. Select OK as required for startup messaging, Open a Unix shell and type `{ctuser@hostname} st` to startup the Application software.

SL #3: Section 7.5 VRAC Flash Update command fails

Issue: `{ctuser@hostname} vrac_flash_update` command fails to pass repeatedly

Solution: Perform the following:

1. Visually verify each VIG Node has a label on the front identifying the hardware as VIG.
2. All Host and VDARC software must be loaded in the correct sequence per the LFC procedure.
3. Verify the correct number of VIG Nodes (1 or 3) present in the Operator Console have been selected during the LFC.

{ctuser@hostname} cat /usr/g/config/INFO

4. With Application software down, verify the security shell for the VDARC Node is successful.

If ssh issues exist, the Operator Console will crash or halt when Application Software is started (startup or st). The message in the gesys log refers to an exit status 9 error. This issue can also present itself as a port 22 error. Other messaging may incorrectly state that the rsh darc failed. The ssh darc command failed.

Gesyslog error message:

NOTE: *Tue Jul 18 09:18:59 2006 Error: 290014004*

Host: ct02

Process: gre-verify-darc

File: gre-verify-darc.cpp

Line: 179

Unable to verify DARC system startup.

/usr/g/bin/gre-verify-darc: failure - waiting for rsh darc response (exit status: 9)

{ctuser@hostname} ssh darc

Last login: Wed May 23 09:55:06 from oc

[ctuser@darc ~] exit

If ssh darc fails:

[root@hostname] rm -rf /usr/g/ctuser/.ssh/known_hosts

[root@hostname] config_sshd

Verify ssh darc now works:

{ctuser@hostname} ssh darc

Last login: Wed May 23 09:55:06 from oc

[ctuser@darc ~] exit

If any other ssh issues exist perform the following:

{ctuser@hostname} **rsh darc**

Last login: Wed May 23 09:55:06 from oc

[ctuser@darc ~] **su -**

Password: **#bigguy**

[root@darc] **service sshd start**

[root@darc] **chkconfig sshd on**

Verify ssh darc now works:

{ctuser@hostname} **ssh darc**

Last login: Wed May 23 09:55:06 from oc

[ctuser@darc ~] **exit**

5. Cycle the VIG Node(s) power on the front panel of each VIG Node present in the Operator Console.

{ctuser@hostname} **rsh darc**

Last login: Wed May 23 09:55:06 from oc

[ctuser@darc ~] **vrac_flash_update ig1**

[ctuser@darc ~] **vrac_flash_update ig2**

[ctuser@darc ~] **vrac_flash_update ig3**

6. After performing sanity scans, the user must halt the Operator Console and then manually power the Operator Console to verify the VIG Node(s) come up properly without any Error messages.

SL #4: Service Software section is not clear enough for the user

Issue: Section 8 Service Software requires clarification

Solution: Update the procedure as follows:

8 Service Software

Service software (Class C) applies only to GE Healthcare personnel and customers with an Advance Service Limited License agreement. Load service software per the instructions on the media.

1. Open a Unix Shell and type:

NOTE:

The Class C and Class M software version (example xxMWxx.x) must match the software set loaded. See Section 2, Software Deliverables.

2. {ctuser@hostname} su -
3. Password: <Password>
4. Insert Class C service software into the Host Computer DVD drive tray.
5. Type: [root@hostname] install classC
6. To close the "sh" window once Class C install is complete, place the cursor in the original terminal window and press [Ctrl]+[C] or double-click on upper left corner of "sh" window.
7. Remove Class C service software from the Host Computer DVD drive and close tray.
8. (For GE Healthcare personnel only) Install Class M service software.

SL #5: A reboot is required after patch installation

Issue: Section 20 Finalization - patch installation sub-steps require a reboot if any patches are installed

Solution: add step 13f. If any patches were installed perform the following:

[root@hostname] **reboot**

Software General Issues:

SW #1: VDARC Node dmidecode command may not always work

Issue: If the VDARC Node does not have software (Linux OS and VDARC Application) loaded on it, the dmidecode command cannot be used to identify the VDARC Node type.

Solution: Instead of using dmidecode to view the type of VDARC Node hardware present in the GOC5 Operator Console, use the ipmitool. Visually verify the output:

JR (Jarrell) or WV (Westville) in Board Product. The following is a Jarrell type example:

```
{ctuser@hostname} ipmitool -H darc -P "fru" | grep -i "Board Product"
```

Board Product : SE7520JR23

Board Product : Intel(R) Management Module - Professional Edition

SW #2: Software Download / InSite Update File Ownership

NOTE:

This must be done every time a LFC is performed

To enable usage of Update Management buttons in Service Desktop/Utilities, the ownership of three files listed below must be set at this time as ctuser.

1. Open unix shell and verify that you are ctuser (not as root).
2. Type **touch /tmp/cl** <Enter>
3. Type **touch /tmp/cl1** <Enter>
4. Type **touch /tmp/cl2** <Enter>

5. To verify that ownership for cl, cl1, cl2 is now ctuser!

Type **cd /tmp** <Enter>

Type **ls -al cl*** <Enter>

Output should look similar to below, note the ownership is ctuser

```
-rw-rw-rw- 1 ctuser users 9 May 31 11:38 cl
-rw-rw-rw- 1 ctuser users 2 May 31 11:38 cl1
-rw-rw-rw- 1 ctuser users 27 May 31 11:38 cl2
-rw-rw-rw- 1 insite root 70 Jun 1 07:00 clientdrv.trace
```

NOTE:

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